

Linear Regression Model for House Price Prediction

Introduction

Machine Learning is a branch of Artificial Intelligence that allows systems to learn patterns from data and make predictions. In this project, a Linear Regression model is used to predict house prices using the California Housing dataset.

The objective of this project is to understand the complete Machine Learning workflow, including:

- Data loading
- Data exploration
- Data visualization
- Data preprocessing
- Model training
- Model evaluation

Python libraries such as pandas, NumPy, matplotlib, seaborn, and scikit-learn were used to implement the project.

Dataset Description

The California Housing dataset is available in the scikit-learn library. It contains information collected from California districts.

The dataset includes several features such as:

- MedInc → Median income
- HouseAge → Average house age
- AveRooms → Average number of rooms
- AveBedrms → Average bedrooms
- Population → Population in the area
- AveOccup → Average occupancy
- Latitude → Geographic latitude

- Longitude → Geographic longitude

Target Variable:

- Price → Median house value

The dataset contains 20,640 rows and 9 columns including the target variable.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify relationships between features.

Steps Performed

1. Dataset Information

The structure of the dataset was checked using:

```
print(df.info())
```

This showed:

- Total number of rows and columns
- Data types of each feature
- No missing values in the dataset

2. Statistical Summary

The dataset statistics were analyzed using:

```
print(df.describe())
```

This provided:

- Mean
- Standard deviation
- Minimum and maximum values
- Quartile values

3. Missing Value Check

The dataset was checked for null values using:

```
print(df.isnull().sum())
```

No missing values were found.

4. Correlation Heatmap

A heatmap was generated using seaborn to visualize feature relationships.

Observations:

- Median income had a strong positive correlation with house prices.
 - Latitude and longitude showed negative correlation with price.
 - Most features showed moderate relationships.
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Model Building

Train-Test Split

The dataset was divided into training and testing sets.

- 80% data used for training
- 20% data used for testing

```
X_train, X_test, y_train, y_test = train_test_split(  
X, y, test_size=0.2, random_state=42  
)
```

Linear Regression Model

A Linear Regression model was created using scikit-learn.

```
model = LinearRegression()  
model.fit(X_train, y_train)
```

The model learned the relationship between housing features and house prices.

Prediction and Evaluation

After training, predictions were made on the testing dataset.

```
y_pred = model.predict(X_test)
```

The model performance was evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 Score

Evaluation Metrics

Mean Absolute Error (MAE)

MAE measures the average prediction error.

Root Mean Squared Error (RMSE)

RMSE measures how far predictions are from actual values.

R^2 Score

R^2 score shows how well the model explains the variance in the dataset.

Higher R^2 values indicate better model performance.

Visualization of Predictions

A scatter plot of Actual Prices vs Predicted Prices was generated.

Observations:

- Most points followed a linear pattern.
 - Predictions were reasonably close to actual prices.
 - Some outliers were present.
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Technologies Used

- Python
- pandas
- NumPy
- matplotlib
- seaborn

- scikit-learn
 - Visual Studio Code
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Conclusion

This project successfully implemented a Linear Regression model for house price prediction using the California Housing dataset.

The project demonstrated the complete Machine Learning workflow:

- Loading data
- Performing EDA
- Visualizing relationships
- Training a model
- Evaluating performance

The model produced reasonable predictions and achieved good performance.

Future Improvements

The project can be improved further by:

- Using advanced regression models
 - Performing feature engineering
 - Hyperparameter tuning
 - Using larger datasets
 - Deploying the model using a web application
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References

1. Scikit-learn Documentation
2. California Housing Dataset
3. Python Documentation

4. Machine Learning Tutorials